

The empirical recurrence for OEIS A300376, recovered from its tail

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Abstract

OEIS A300376 records “Empirical recurrence of order 80” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 96$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 3$, so the recurrence is proved for every $n \geq 83$. Its last coefficient is $c_{80} = -1 \neq 0$, so no further tail satisfies anything shorter; any order-80 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A300376 is “Number of $n \times 4$ 0..1 arrays with every element equal to 0, 2 or 3 horizontally, vertically or antidiagonally adjacent elements, with upper left element zero.”. It has offset 1 and begins

1, 10, 11, 26, 80, 237, 692, 2548, ...

The entry states:

Empirical recurrence of order 80 (see link above)

As of the “Last modified” line on the live entry (Mar 04 2018 09:17:26, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 96$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 96$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 80: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 3$ is the first whose minimal order is exactly the stated 80.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 192$ exact terms from $n = 3$ on, it returns order 80 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{80} c_i a(n-i),$$

c_1	0	c_{21}	11947	c_{41}	−690148	c_{61}	25507
c_2	8	c_{22}	9092	c_{42}	665874	c_{62}	−26118
c_3	16	c_{23}	−32355	c_{43}	365210	c_{63}	12229
c_4	8	c_{24}	−33021	c_{44}	−1303531	c_{64}	−3204
c_5	−29	c_{25}	−11148	c_{45}	613455	c_{65}	−781
c_6	−80	c_{26}	−59213	c_{46}	−450321	c_{66}	3818
c_7	−145	c_{27}	22871	c_{47}	−340901	c_{67}	−2011
c_8	−428	c_{28}	2224	c_{48}	776556	c_{68}	341
c_9	−464	c_{29}	129117	c_{49}	−277184	c_{69}	565
c_{10}	781	c_{30}	307030	c_{50}	107976	c_{70}	−545
c_{11}	1410	c_{31}	97366	c_{51}	302160	c_{71}	−95
c_{12}	1785	c_{32}	353177	c_{52}	−453987	c_{72}	160
c_{13}	3857	c_{33}	−270666	c_{53}	191154	c_{73}	32
c_{14}	2025	c_{34}	−266444	c_{54}	−50777	c_{74}	−85
c_{15}	−76	c_{35}	−4731	c_{55}	−145758	c_{75}	41
c_{16}	−464	c_{36}	−1056011	c_{56}	179687	c_{76}	12
c_{17}	−10012	c_{37}	408214	c_{57}	−81065	c_{77}	−23
c_{18}	−13743	c_{38}	−267400	c_{58}	29533	c_{78}	5
c_{19}	−6735	c_{39}	−226438	c_{59}	28767	c_{79}	3
c_{20}	−2892	c_{40}	1479340	c_{60}	−37710	c_{80}	−1

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 83$, and it is the recurrence OEIS A300376 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 3$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{80} - c_1 x^{80-1} - \dots - c_{80}$. Its constant term is $-c_{80} = 1$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of

terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 80 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 80, so $\deg q = 80$, and it is monic. It holds from some index t on. From $\max(t, 3)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 80, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 26 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 80, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -1 .

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-80 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A300376.
- [2] J. L. Massey, Shift-register synthesis and BCH decoding, *IEEE Trans. Inform. Theory* **15** (1969), 122–127.
- [3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)
- [4] M. Kauers and P. Paule, *The Concrete Tetrahedron*, Springer, 2011. (C-finite sequences and their closure properties.)
- [5] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.